



SURF



Performance analysis

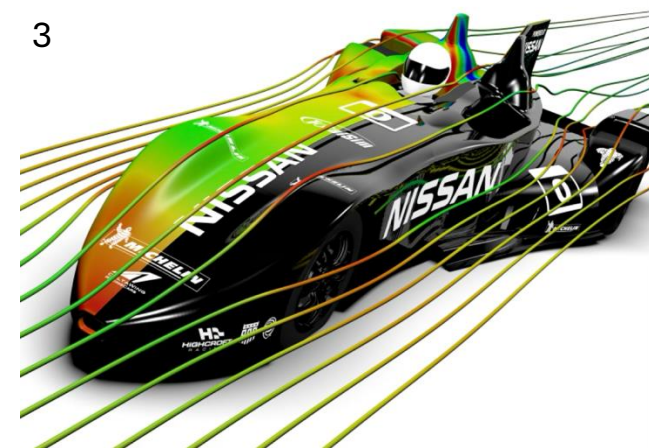
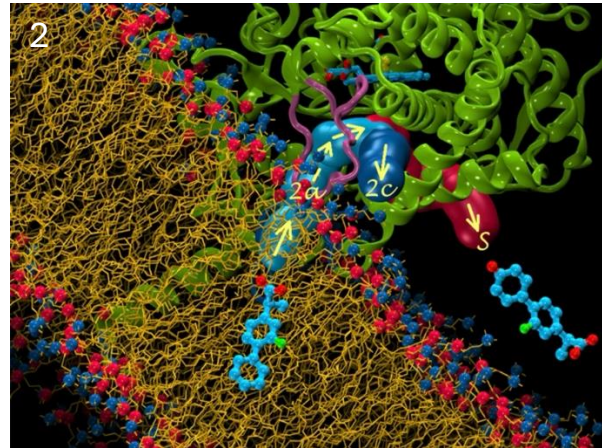
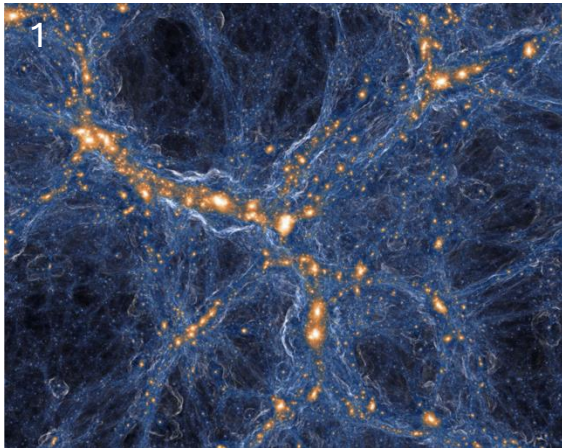
Maxim Masterov

15.05.2025

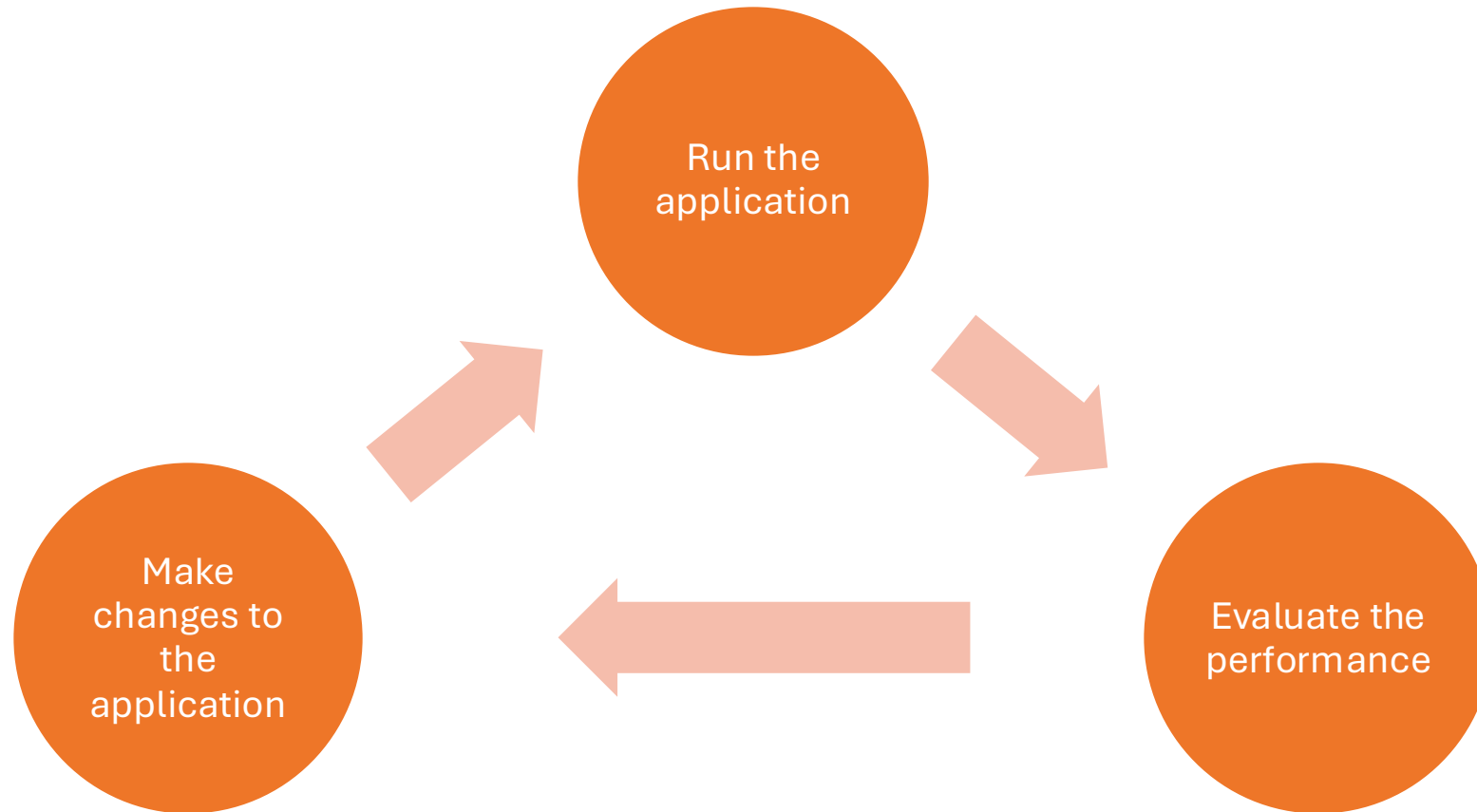
Performance analysis

Overview of HPC applications

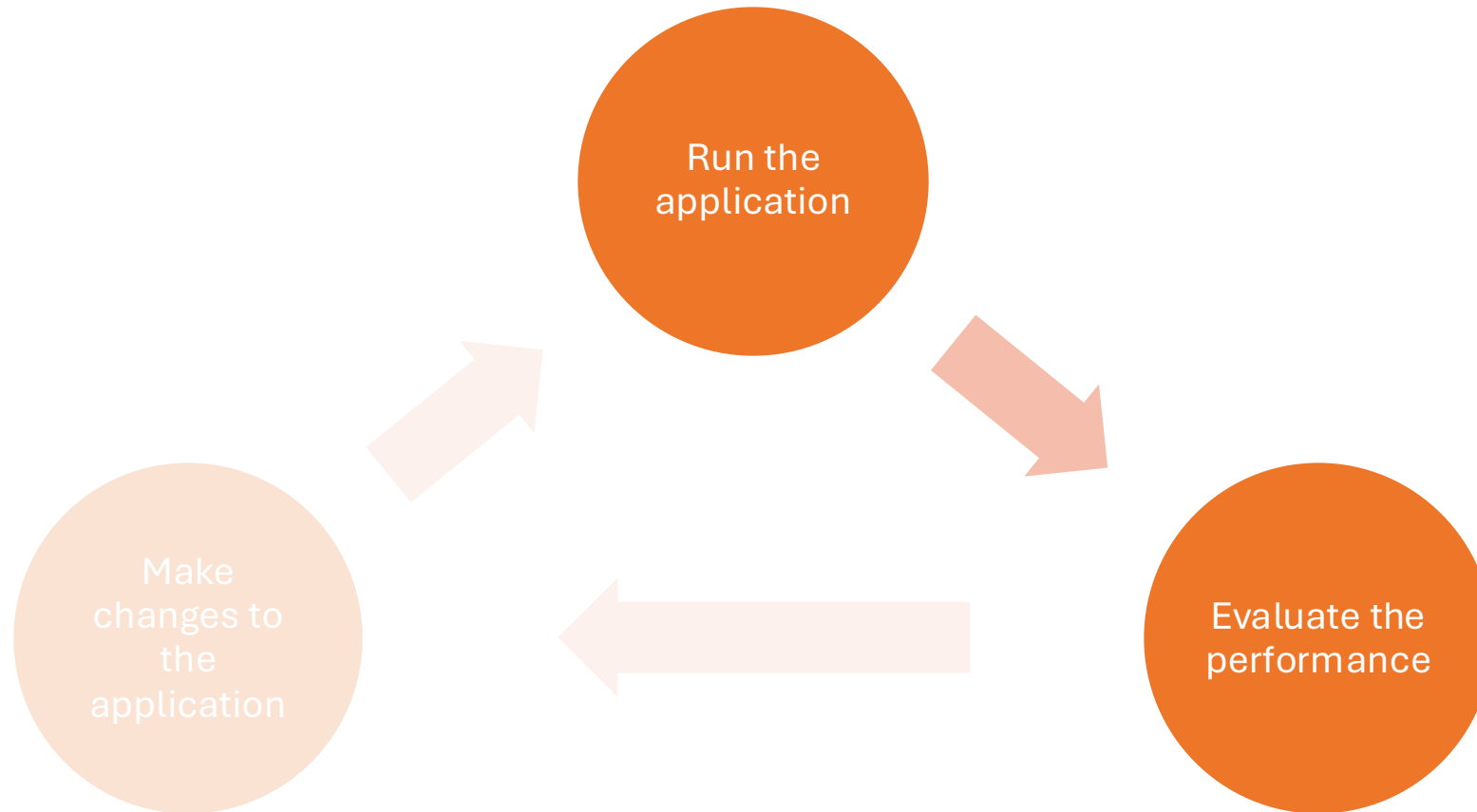
- HPC applications are software designed to exploit the power of supercomputers or high-performance clusters to solve complex problems
- HPC applications perform intensive computations, facilitate communications between parallel processes, and manage input/output operations on a high-performance file system



| A life cycle of an HPC application

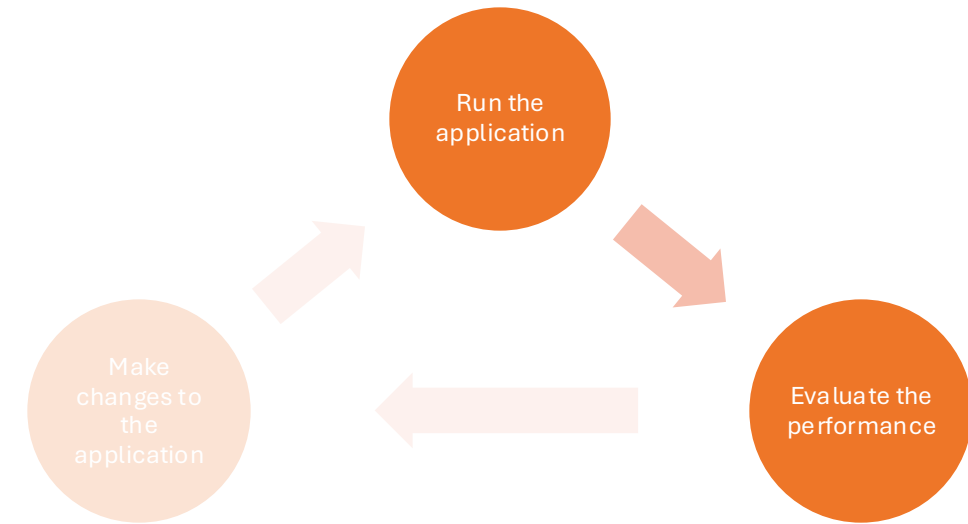


| A life cycle of an HPC application



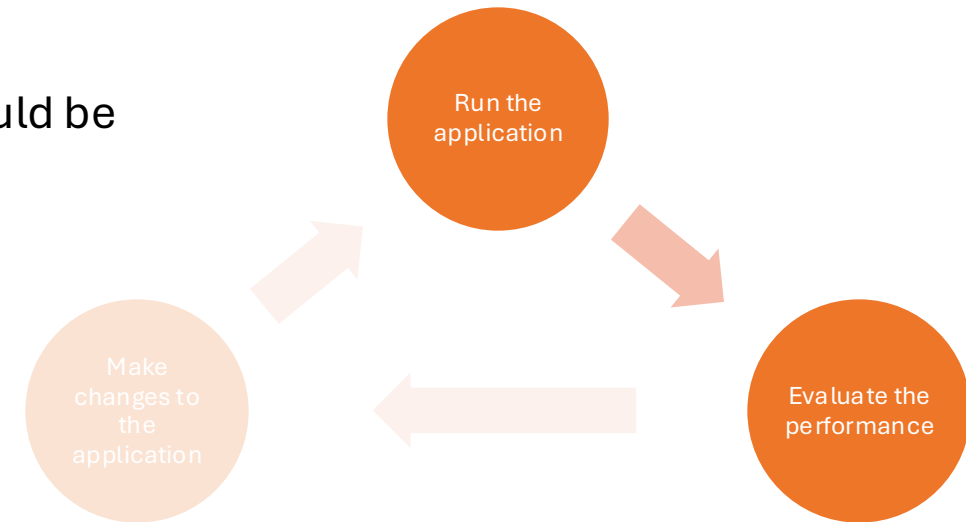
| How to evaluate the performance?

- Using tools, e.g. Vtune, Scalasca, nvprof, memory-profiler
- Measuring the execution time
- Measuring the memory footprint
- Measuring the network load
- Comparing the performance with similar applications/models
- Comparing the performance across different hardware
- Using “guts feeling”



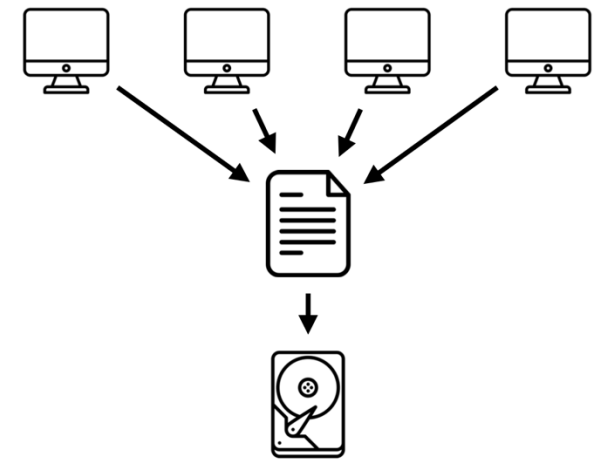
| Performance analysis

- An application consists of multiple levels of complexity
- Each level may have certain effect on the performance and should be evaluated individually:
 - **IO**
 - **Communications**
 - **Computations**
 - **Memory footprint**



| Performance analysis - IO

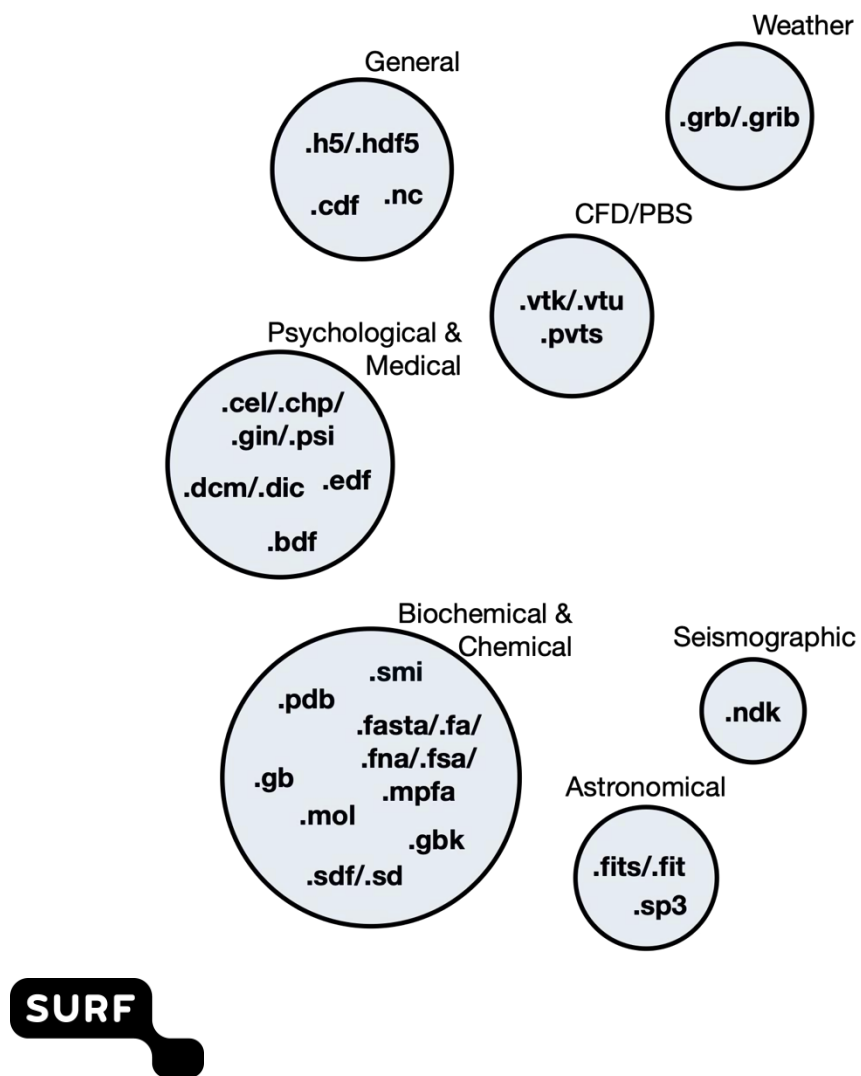
- Parallel vs sequential filesystems
- We can measure read/write speed
- We can use ASCII/Binary
- We can use different file formats, as some are inherently sequential, some are parallel
- The easiest way to evaluate performance is to measure the elapsed time of IO operations



10M vector of 64-bit FP values on NVME file system

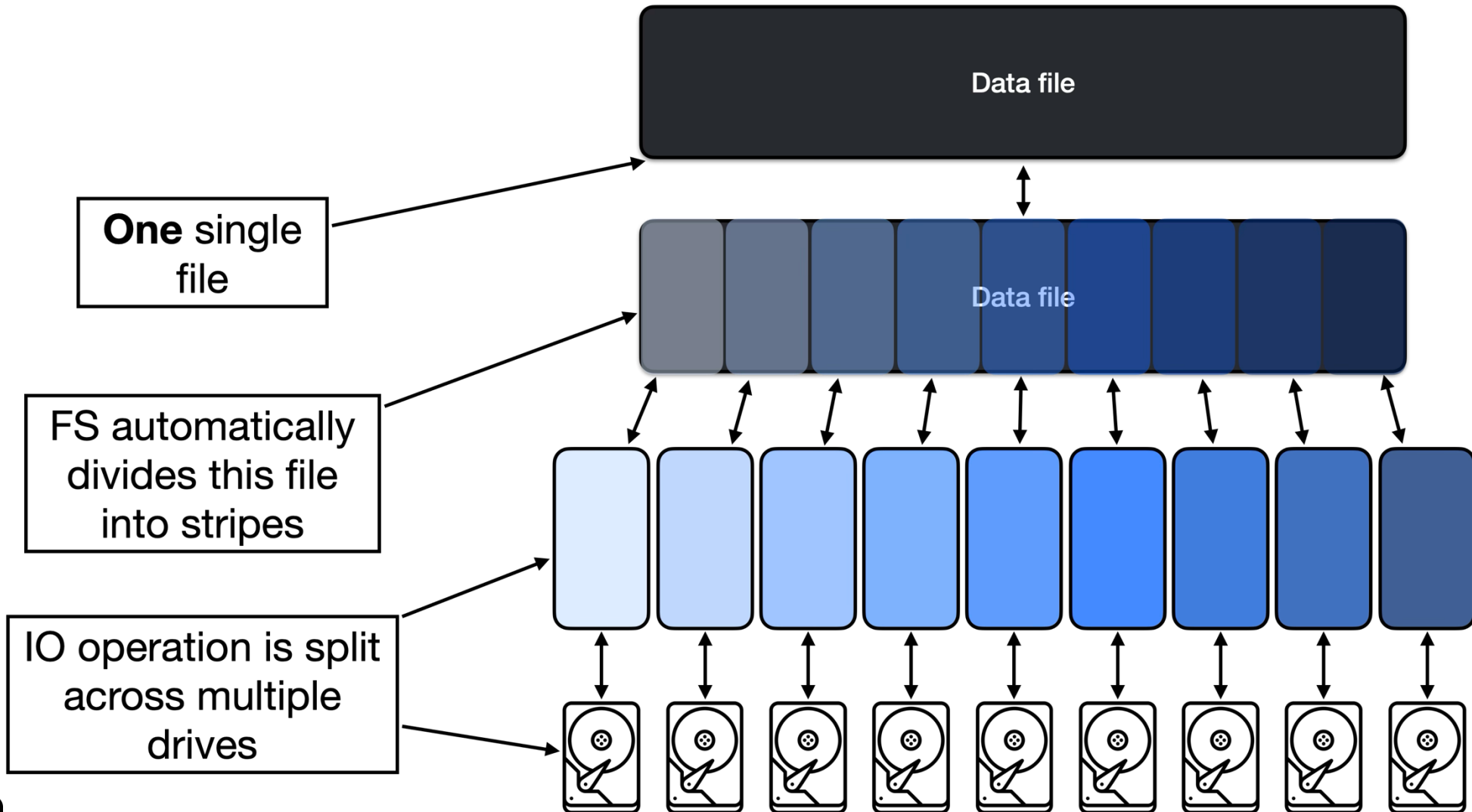
Format	IO operation	Speed, [MB/s]
ASCII	Read	29.7
	Write	24.9
Binary	Read	2666.4
	Write	1586.5

| Performance analysis - IO



IBM
Spectrum
Scale

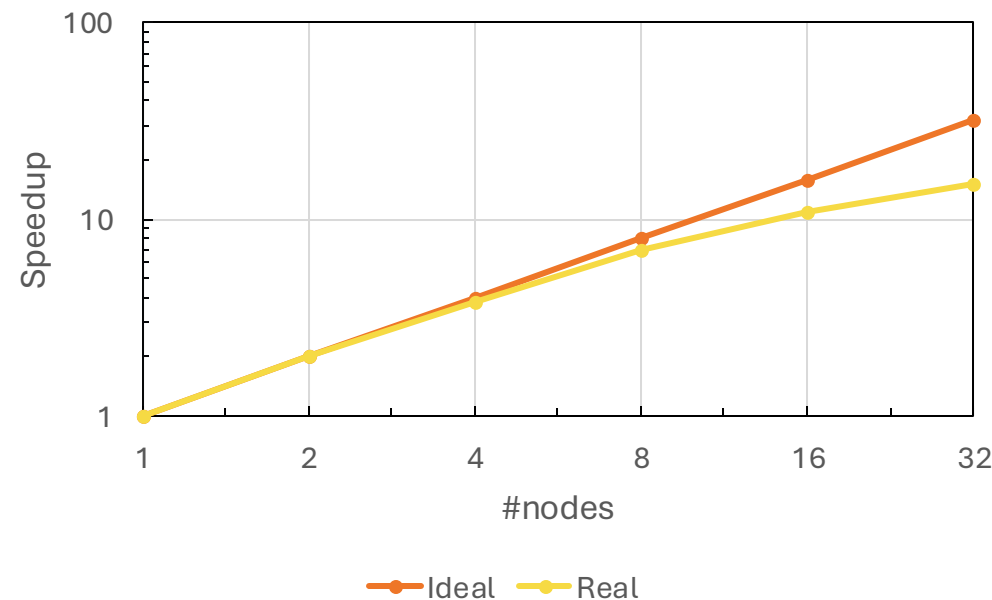
| Performance analysis - IO



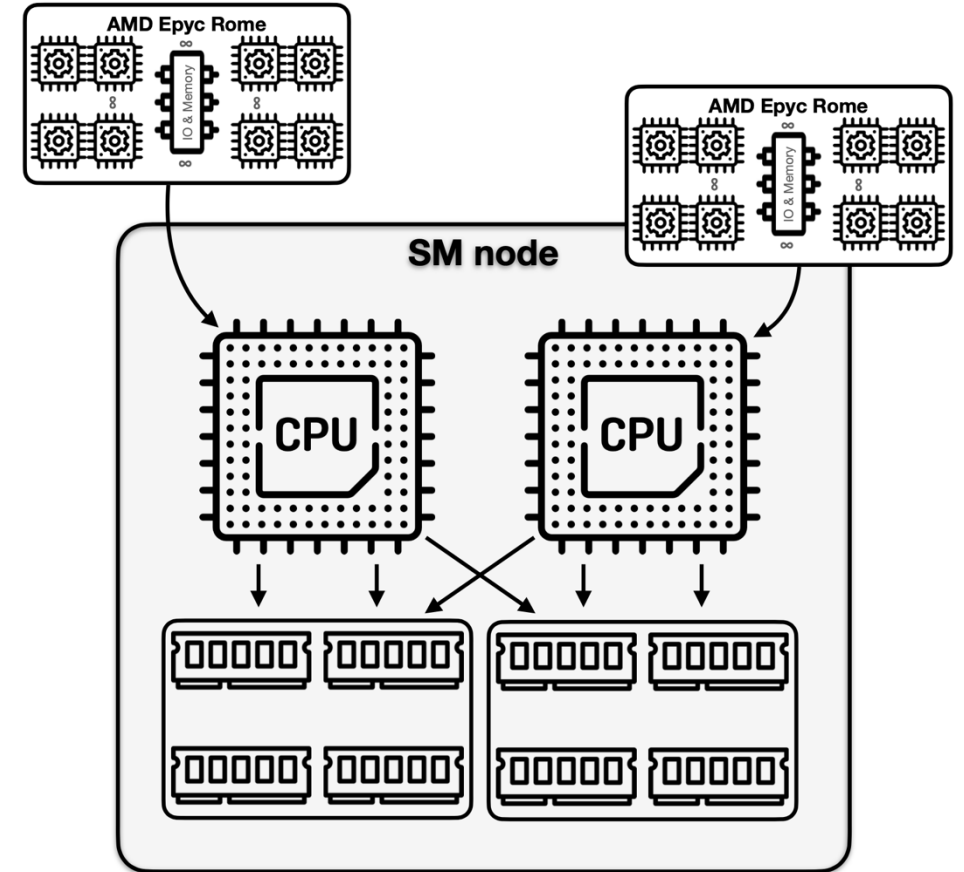
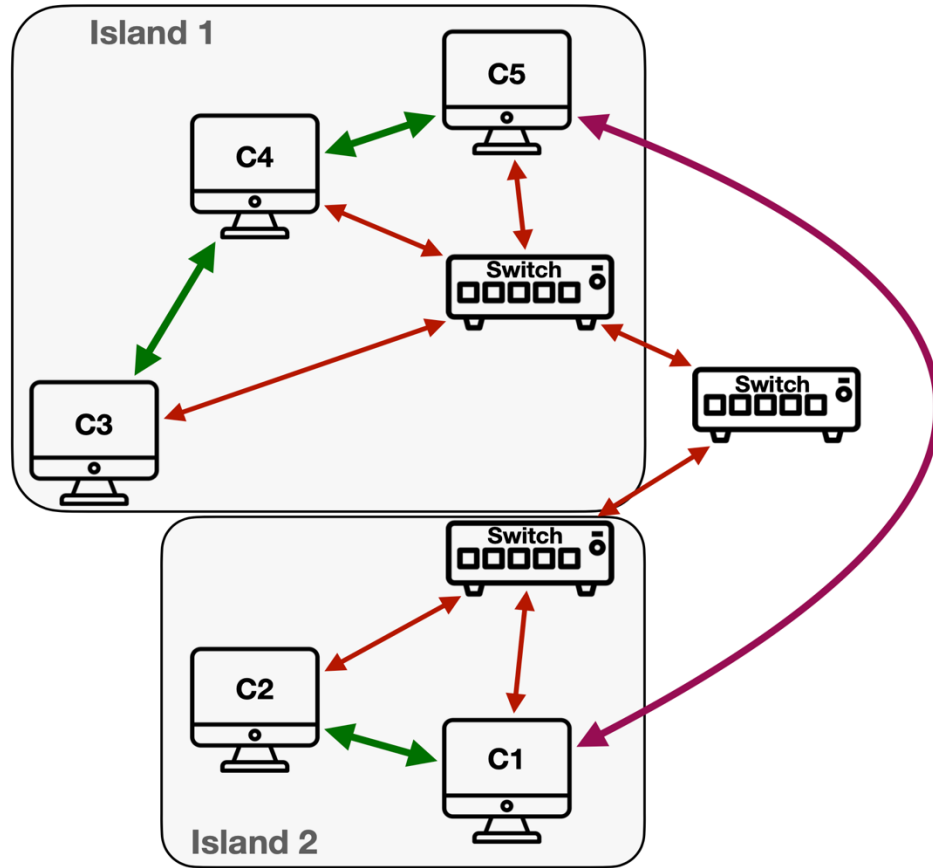
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Performance analysis - Communications

- Leads to a decrease in performance as the number of active processes increases
- Optimisation may require knowledge of the network structure
- Some frameworks allow for run-time tuning of the communication algorithms (e.g. with MPI tune)
- Scales with $O(\sqrt{P})$ for P processes

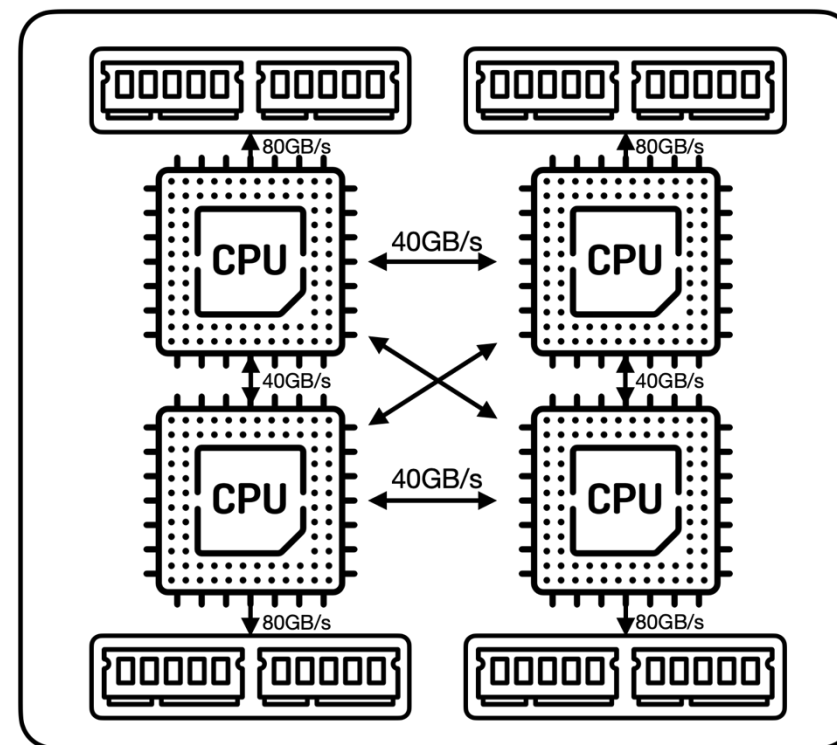
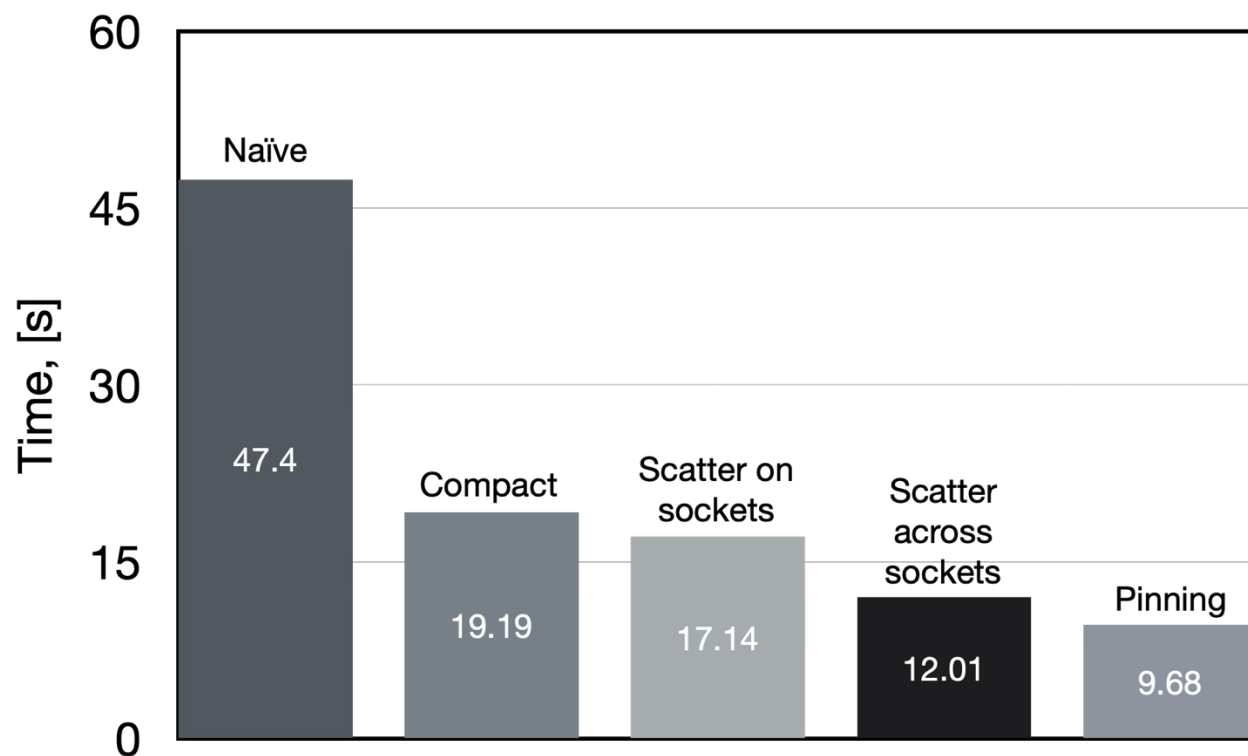


Performance analysis - Communications



Performance analysis - Communications

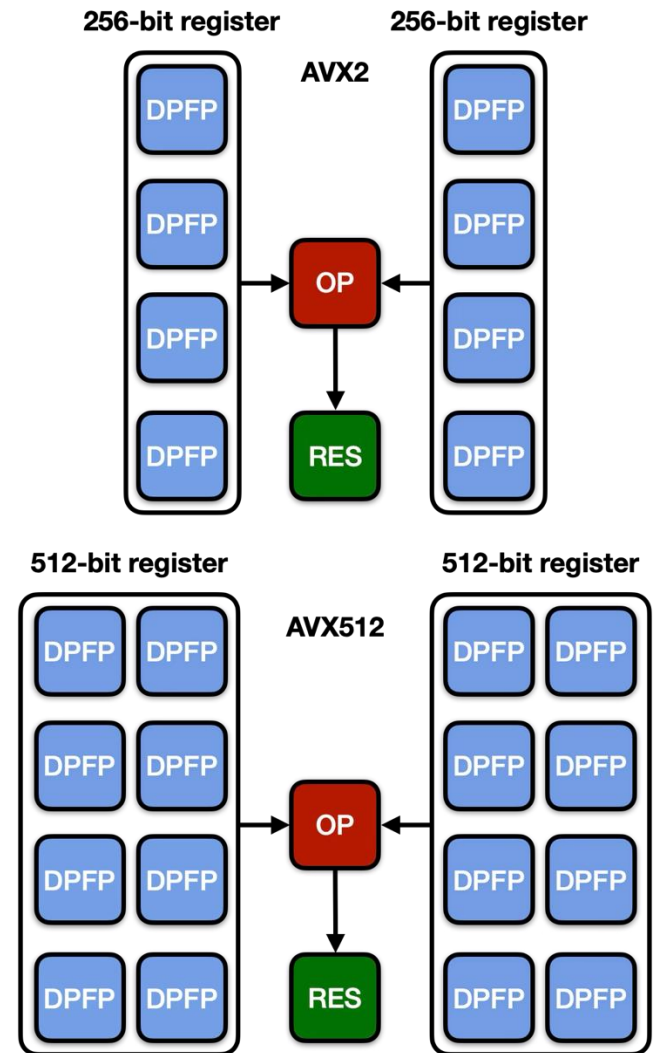
QX-simulator, C++, 32 threads



Performance analysis - Computations

- **Dozens** of packages that allow to analyse performance of sequential and parallel codes
- Different tools suite different parallelisation strategies (threaded, distributed, GPU)
- Many tools use **hardware counters** that **have to be enabled explicitly on Snellius**
- Scales with $O(P)$ for P processes

```
#SBATCH --constraint=hwperf  
#SBATCH --exclusive
```



Performance analysis - Memory

- Memory footprint – how much RAM does an application use
- May affect cache misses and, therefore, performance
- May dictate the scale at which application should be executed

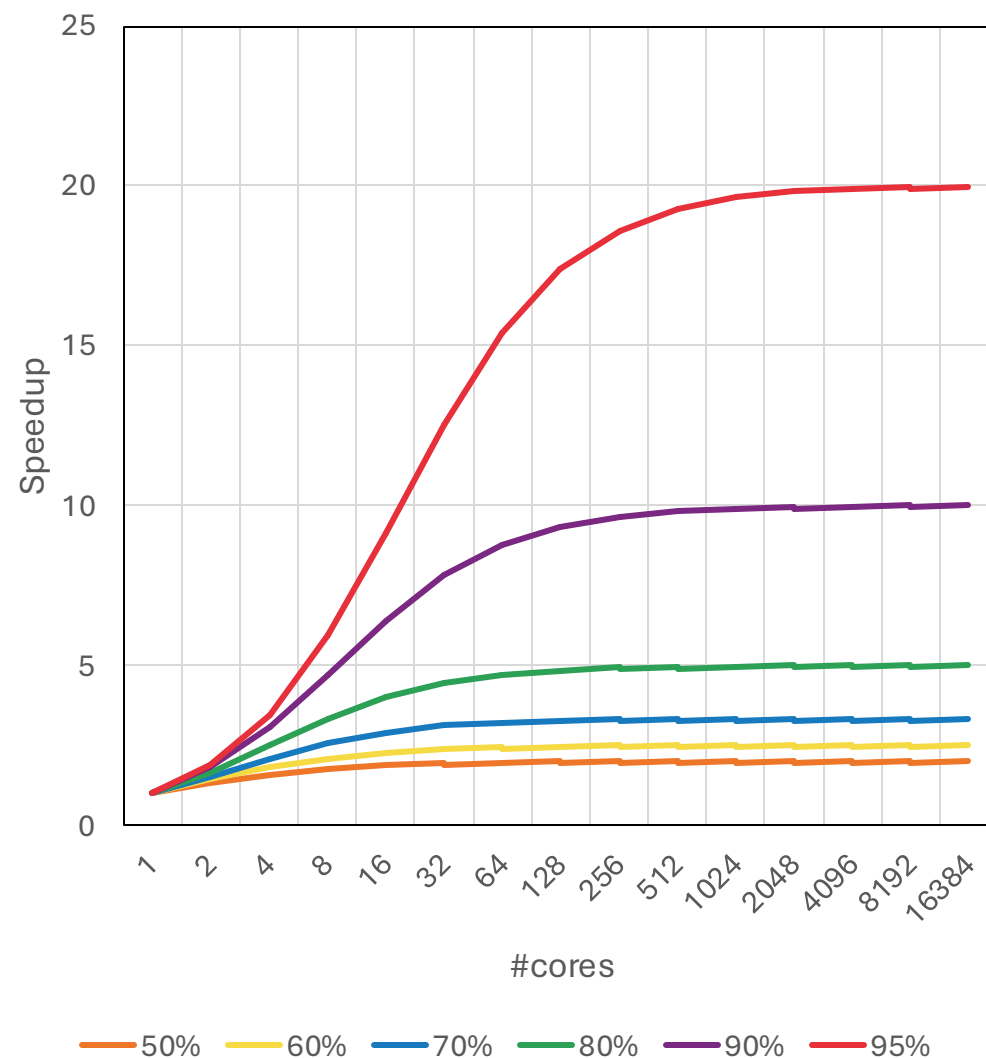
```
$ valgrind --tool=massif <prog>
$ ms_print massif.out.XXX
```

Strong and weak scaling

- **Strong scaling** – the problem size is **fixed**, the number of processors is **increased**
- **Amdahl's law** - the speedup is limited by the fraction of the serial part of the software that is not amenable to parallelization

$$speedup = \frac{1}{s + p/N}$$

- s – proportion of the execution time spent on the serial part
- p – proportion of the execution time spent on the parallel part
- N – number of processes

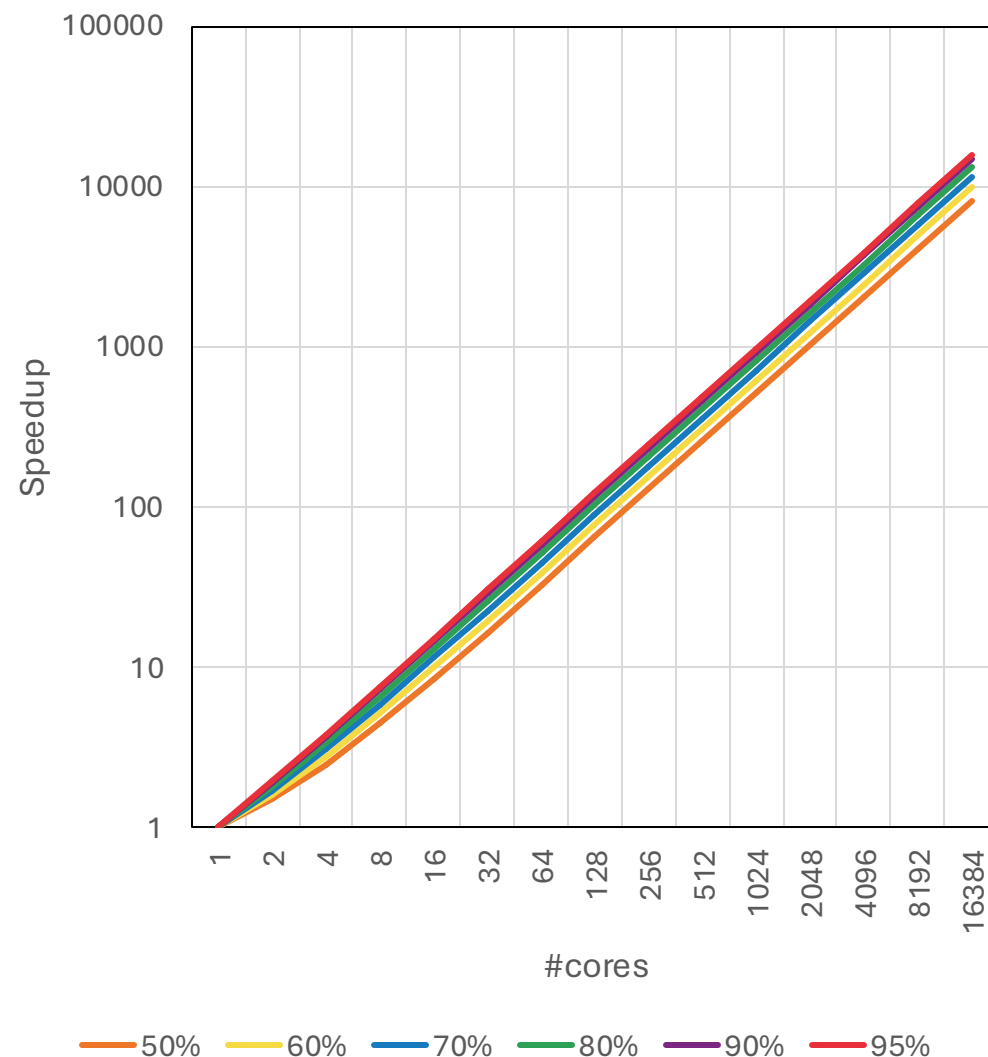


Strong and weak scaling

- **Weak scaling** – the problem size **increases** with the number of processes
- **Gustafson's law** - the parallel part scales linearly with the amount of resources, and the serial part does not increase with respect to the size of the problem

$$speedup = s + p \cdot N$$

- s – proportion of the execution time spent on the serial part
- p – proportion of the execution time spent on the parallel part
- N – number of processes



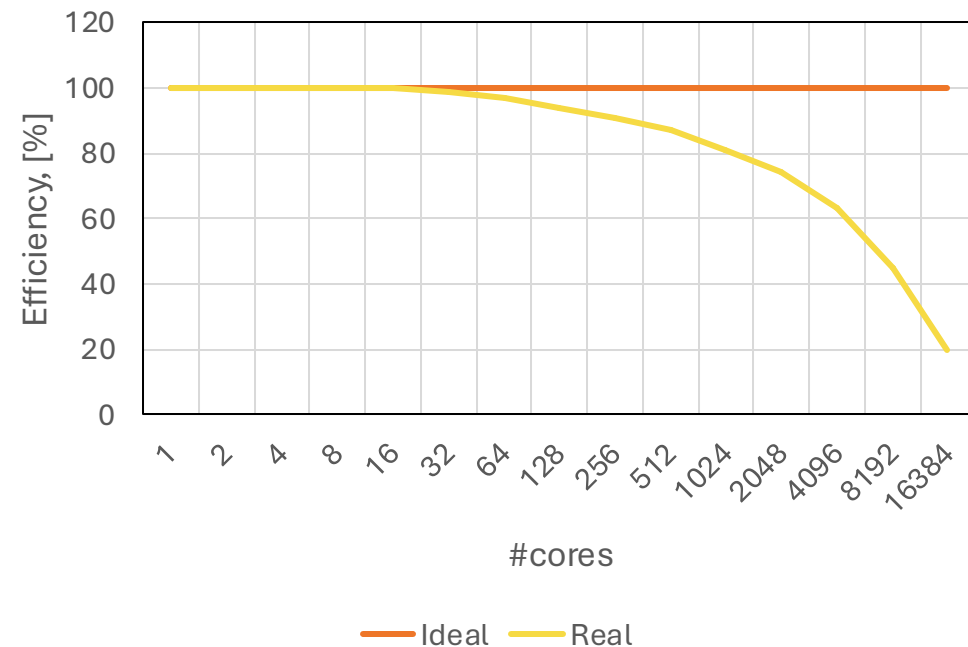
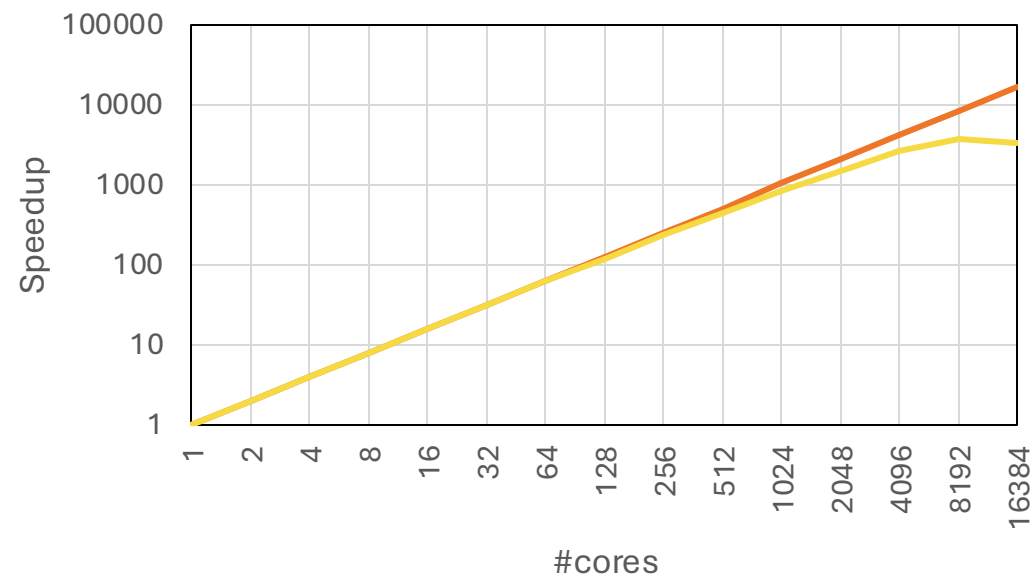
Strong and weak scaling

- Speedup and efficiency:

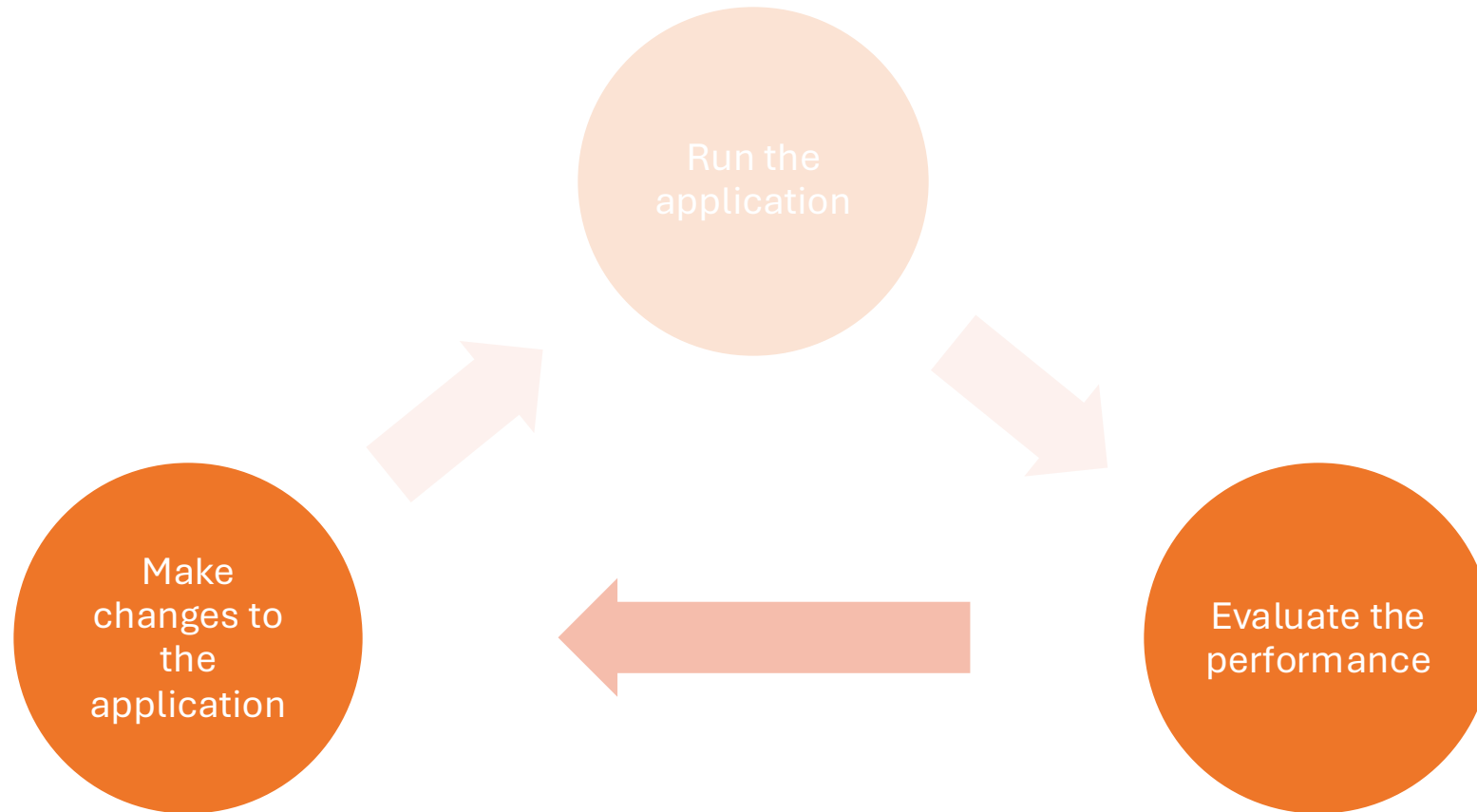
$$speedup = \frac{T_s}{T_N}$$

$$efficiency = \frac{1}{N} \frac{T_s}{T_N}$$

- T_s – the amount of time needed to complete a serial task
- T_N – the amount of time needed to complete a parallel task on N processes
- N – number of processes



| A life cycle of an HPC application

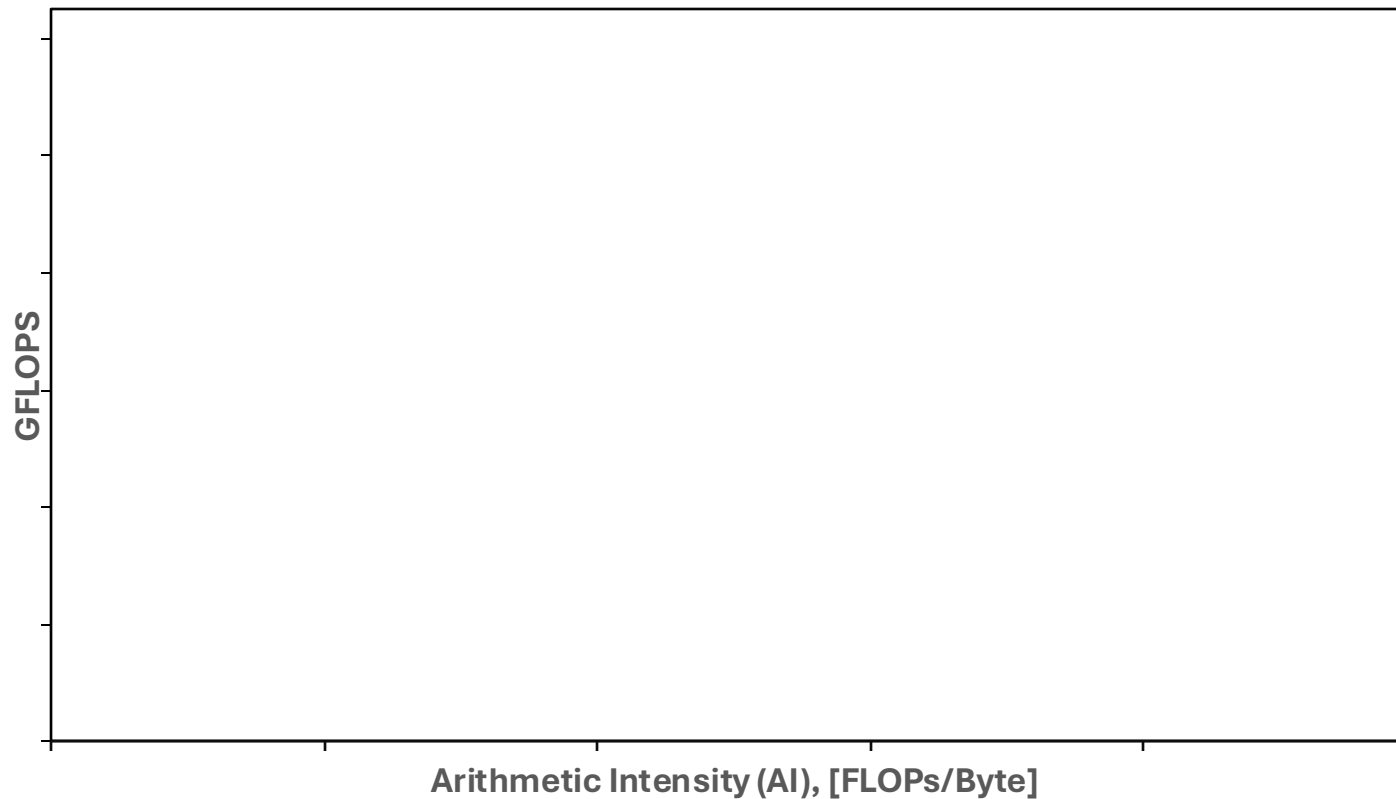


| Standard techniques

- Change the model/method to a faster one
- Change the backend (e.g. linear algebra) to a faster one
- Reduce the precision (e.g. from 64-bit FP to 32-bit FP)
- Optimise data structures (e.g. from AOS to SOA)
- Optimise IO operations
- Optimise data transfers (cache, network)
- Make application hardware-aware
- Use different hardware
- Optimise memory footprint
- Use different programming language

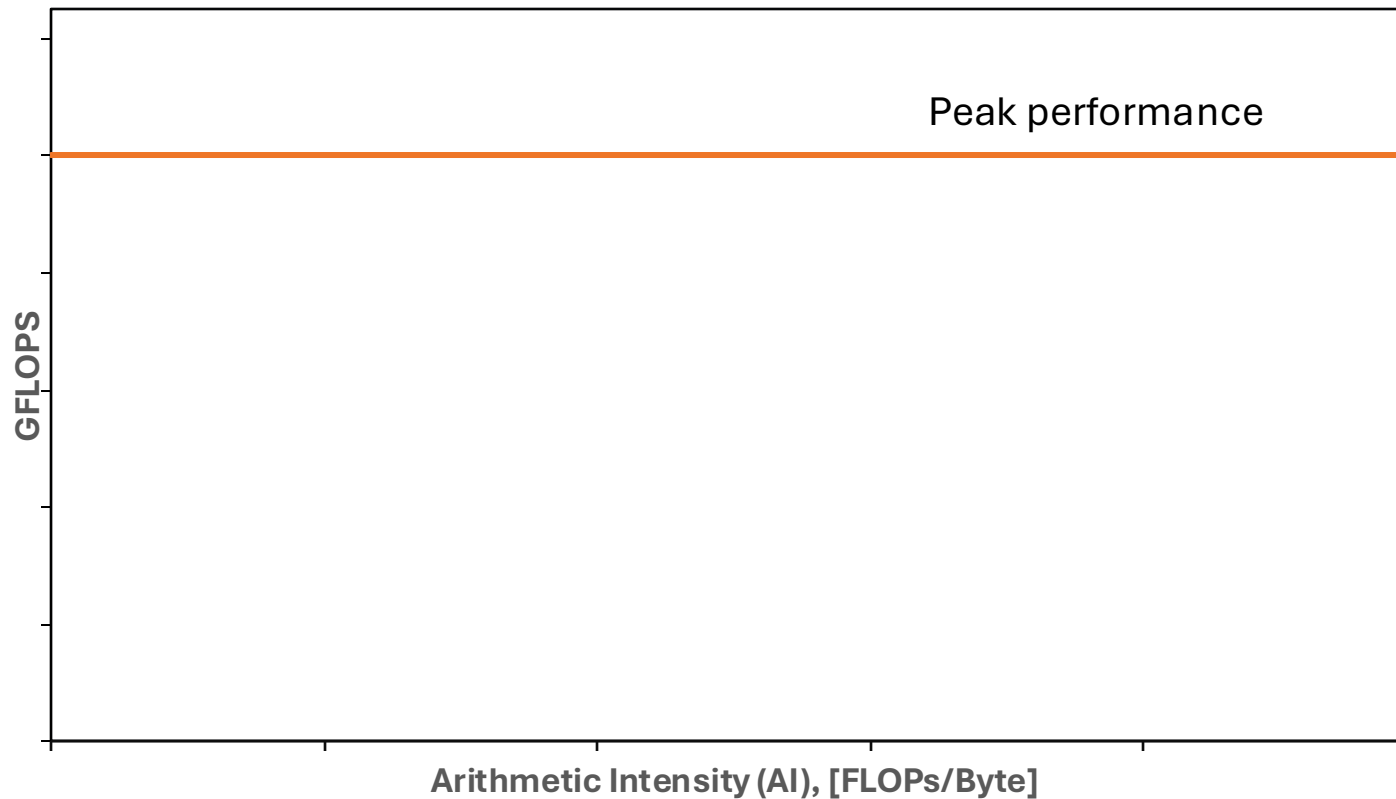
| Roofline model

- A visual model that demonstrates the performance of an application or its kernel with respect to the hardware limitations.
- Demonstrates potential benefits and priority of optimisations.



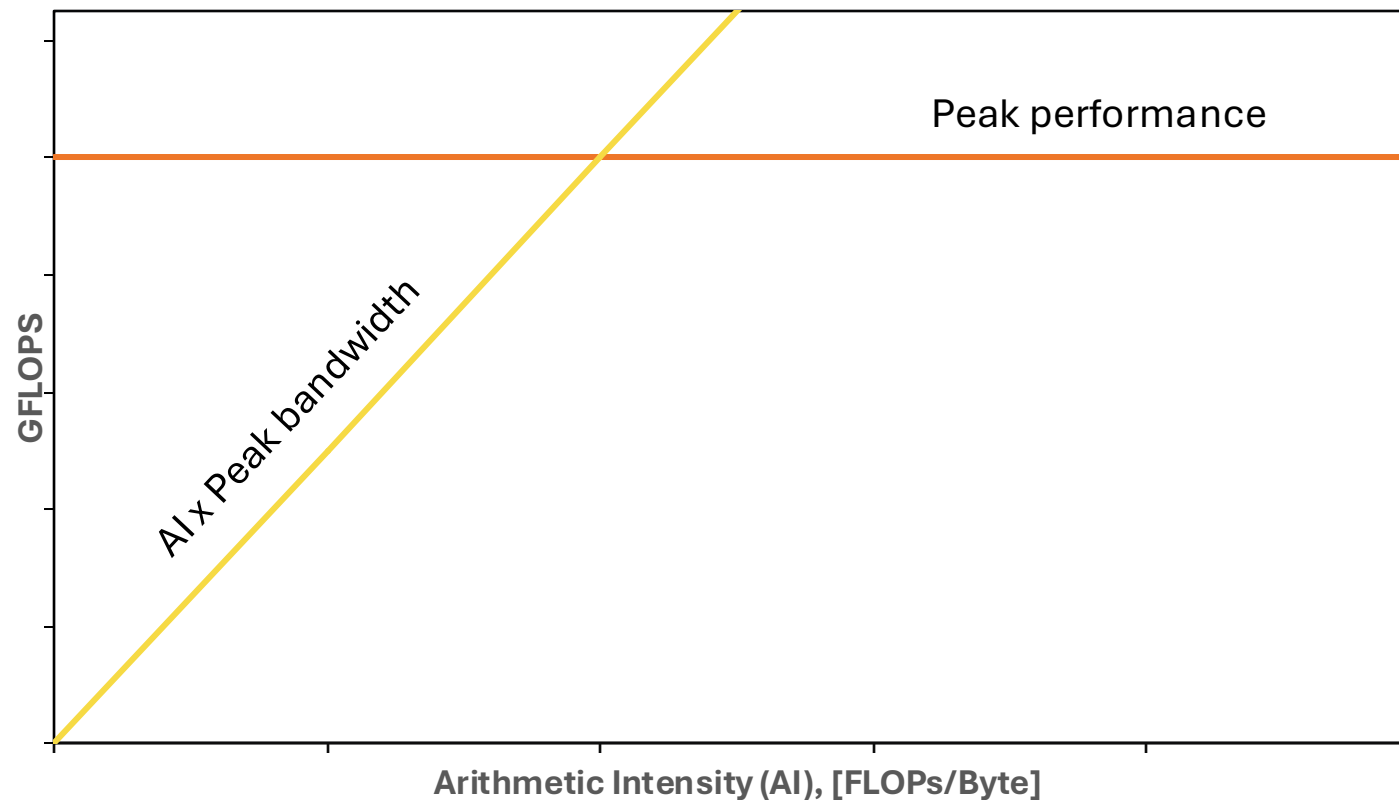
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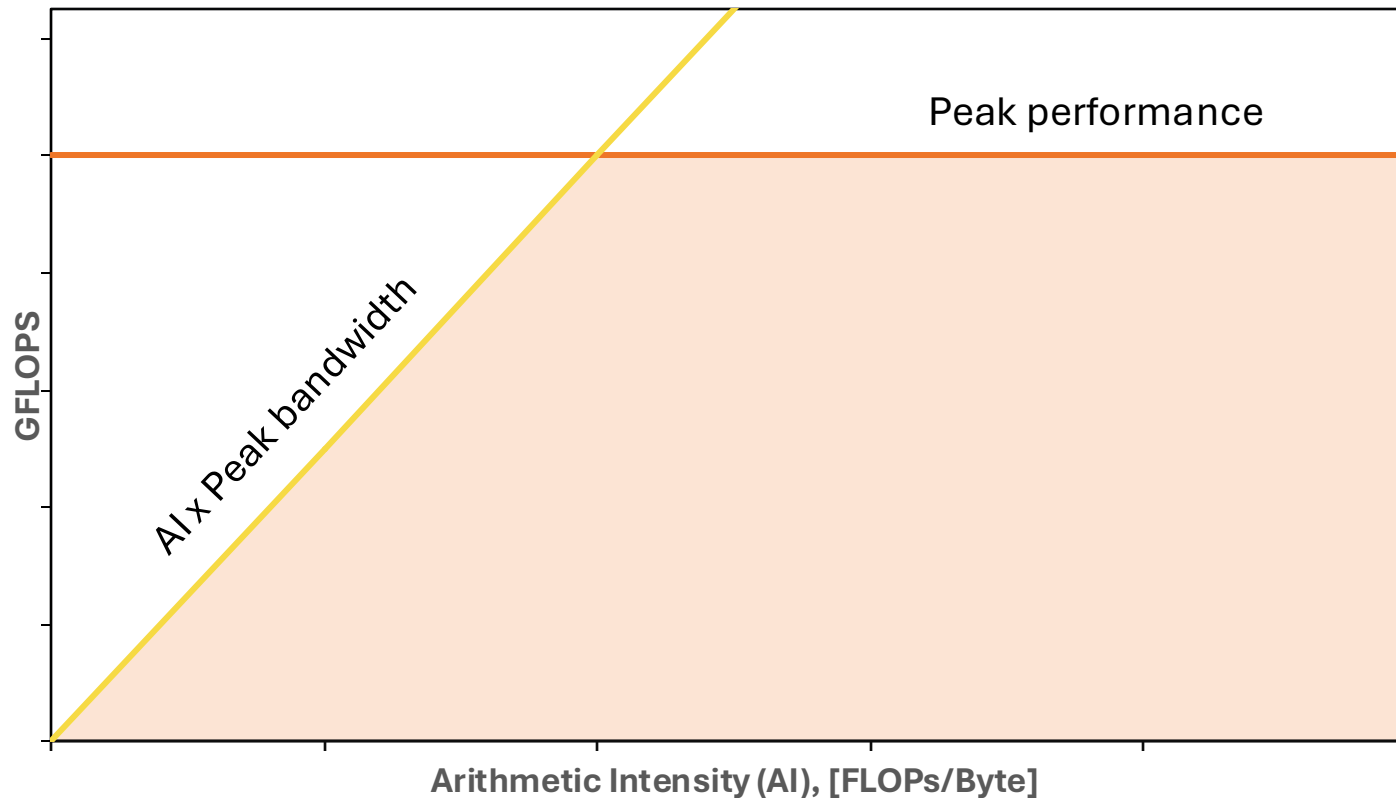
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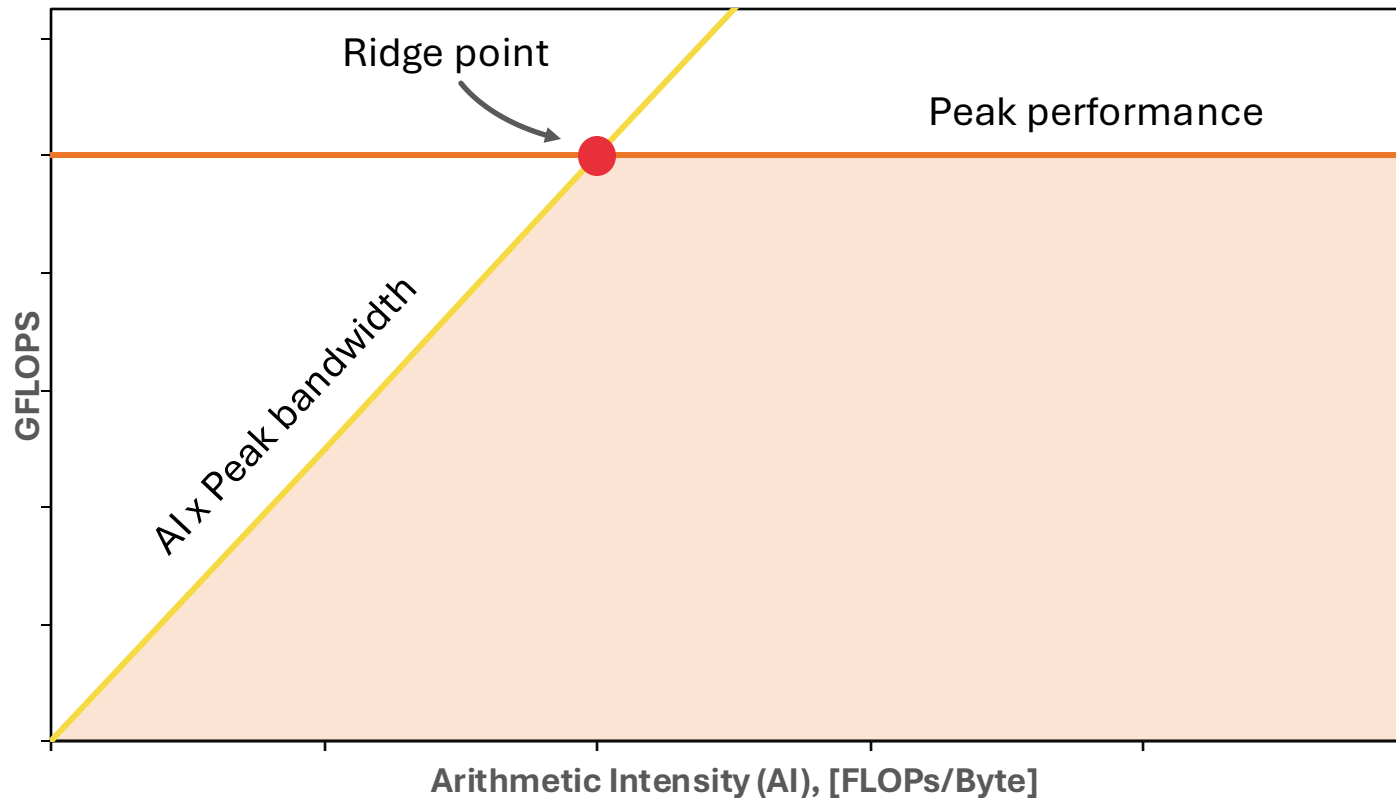
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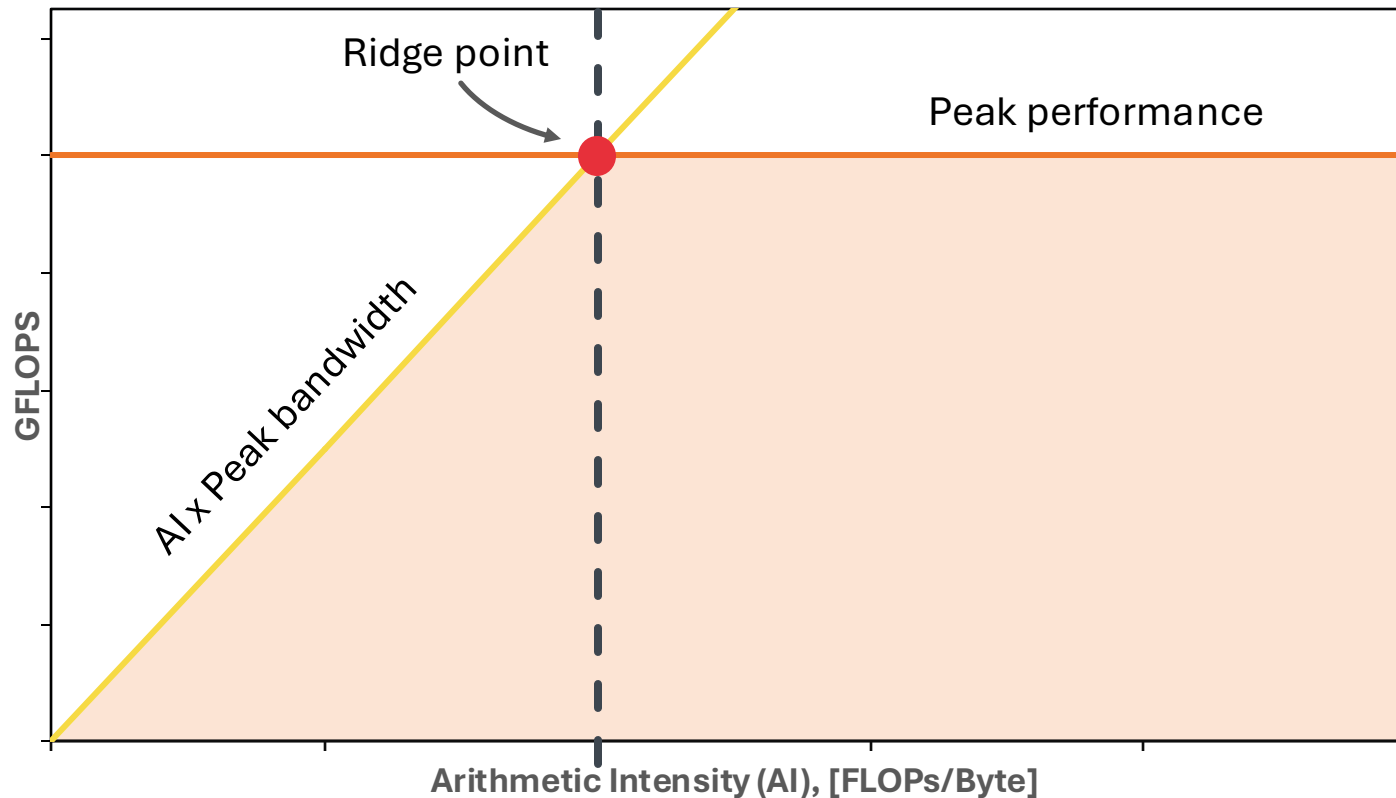
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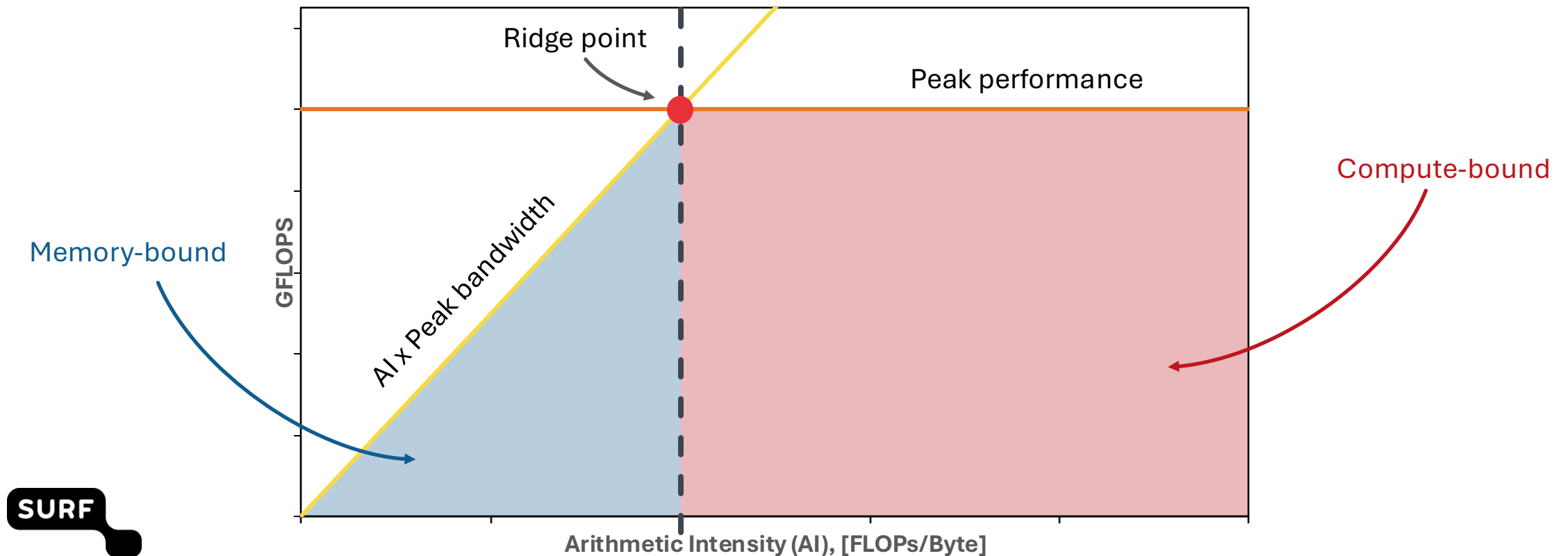
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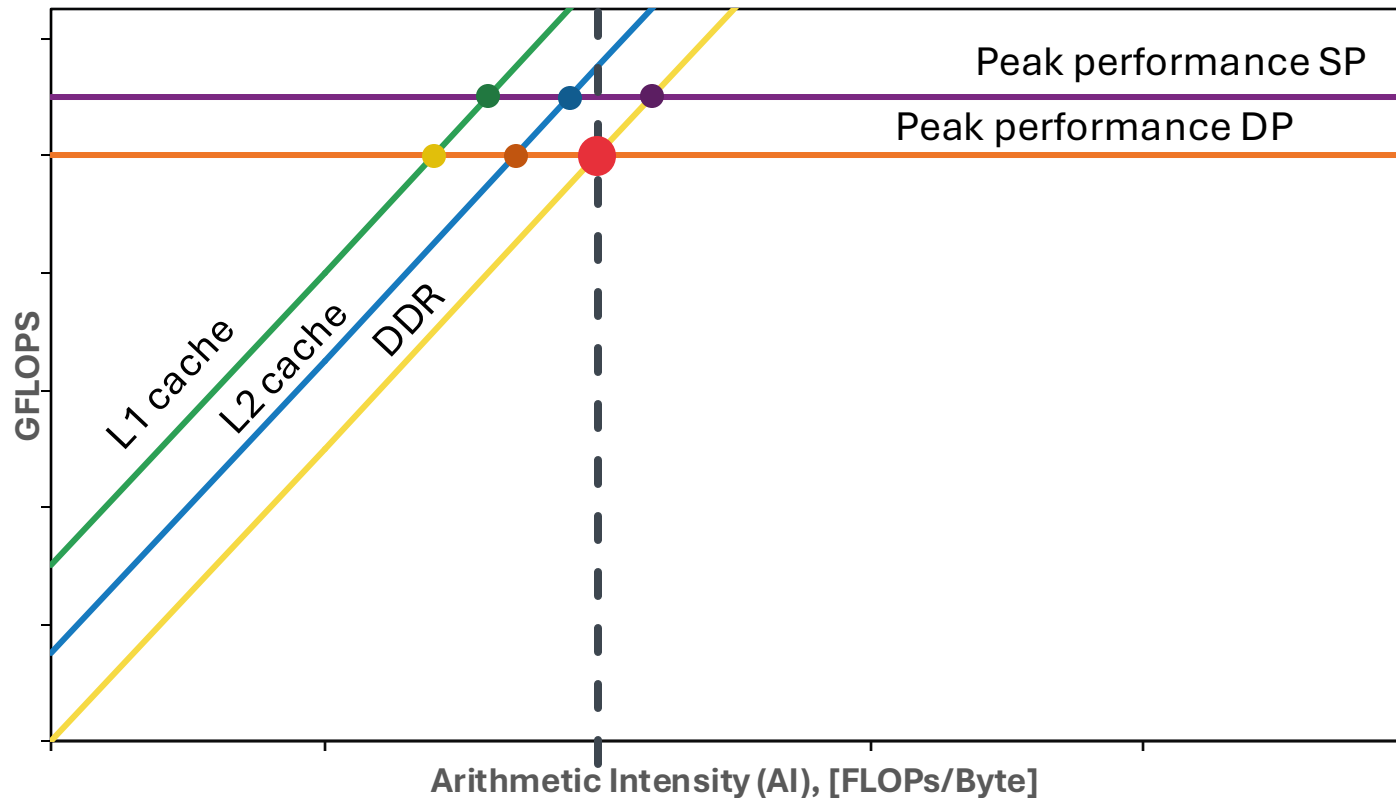
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| How to estimate FLOPs?

- Manual FLOP estimation

```
for n in range(0, N):  
    z[n] = saxpy(a, x, y)
```

```
def saxpy(a, x, y):  
    return a * x + y
```



Addition: 1 FLOP
Multiplication: 1 FLOP

* $N = 2N$ FLOPs

| How to estimate FLOPs?

- Use knowledge of the algorithm complexity (if possible)
- For instance:
 - Generic dense matrix-matrix multiplication: $\approx O(N^3)$ FLOPs
 - FFT algorithm: $\approx O(N \log_2(N))$ FLOPs
- This approach often does not consider the total number of FLOPs in the algorithm

```
def gemm(matA, matB, matC, N):  
    for i in range(0, N):  
        for j in range(0, N):  
            for k in range(0, N):  
                matC[i][j] += matA[i][k] * matB[k][j]
```

$2N^3$ FLOPs

| How to estimate FLOPs?

- What if we have a large code base of complex implementation?

```
def check_collisions_v4():
    tree = cKDTree(positions)
    pairs = np.array(list(tree.query_pairs(radii.max() * 2)))

    if len(pairs) > 0:
        i, j = pairs[:, 0], pairs[:, 1]
        r_rel = positions[i] - positions[j]
        v_rel = velocities[i] - velocities[j]
        # dists = np.linalg.norm(r_rel, axis=1)
        mask = np.einsum('ij,ij->i', v_rel, r_rel) < 0
        valid_pairs = pairs[mask]

        for i, j in valid_pairs:
            resolve_collision(i, j)
```

| How to estimate FLOPs?

- What if we have large code base of complex implementation?
- **Use performance analysis tools!**

Tool name	Costs	Description
ARM DDT	Non-free	Full featured graphical, parallel debugger
HPCToolkit	Free	Integrated suite of tools for parallel program performance analysis
Intel One API	Free under certain conditions	Stack of different performance analysis and debugging tools (MPI/OpenMP/SIMD)
Valgrind	Free	Memory errors debugging tool
TotalView	Non-free	Full featured graphical, parallel debugger
Vampir	Non-free	Full featured trace visualizer for parallel program OTF trace files
memP	Free	Lightweight memory profiling tool
mpiP	Free	Lightweight MPI profiling tool
PAPI	Free	A standardized and portable API for accessing performance counter hardware
likwid	Free	A tool to measure hardware performance counters
Tau	Free	Full featured parallel program performance analyses toolkit
Extrae	Free	MPI/OpenMP profiler
Scalasca	Free	performance analysis tool for MPI+OpenMP
Darshan	Free	IO profiler
nvprof	Free	Thread profiler (inc. GPU) from NVIDIA
ARM MAP	Non-free	performance analysis tool for MPI+OpenMP
uProf	Free	performance analysis tool for MPI+OpenMP
gprof	Free	Standard unix/linux profiling utility

| How to estimate FLOPs?

- AMD-uProf:

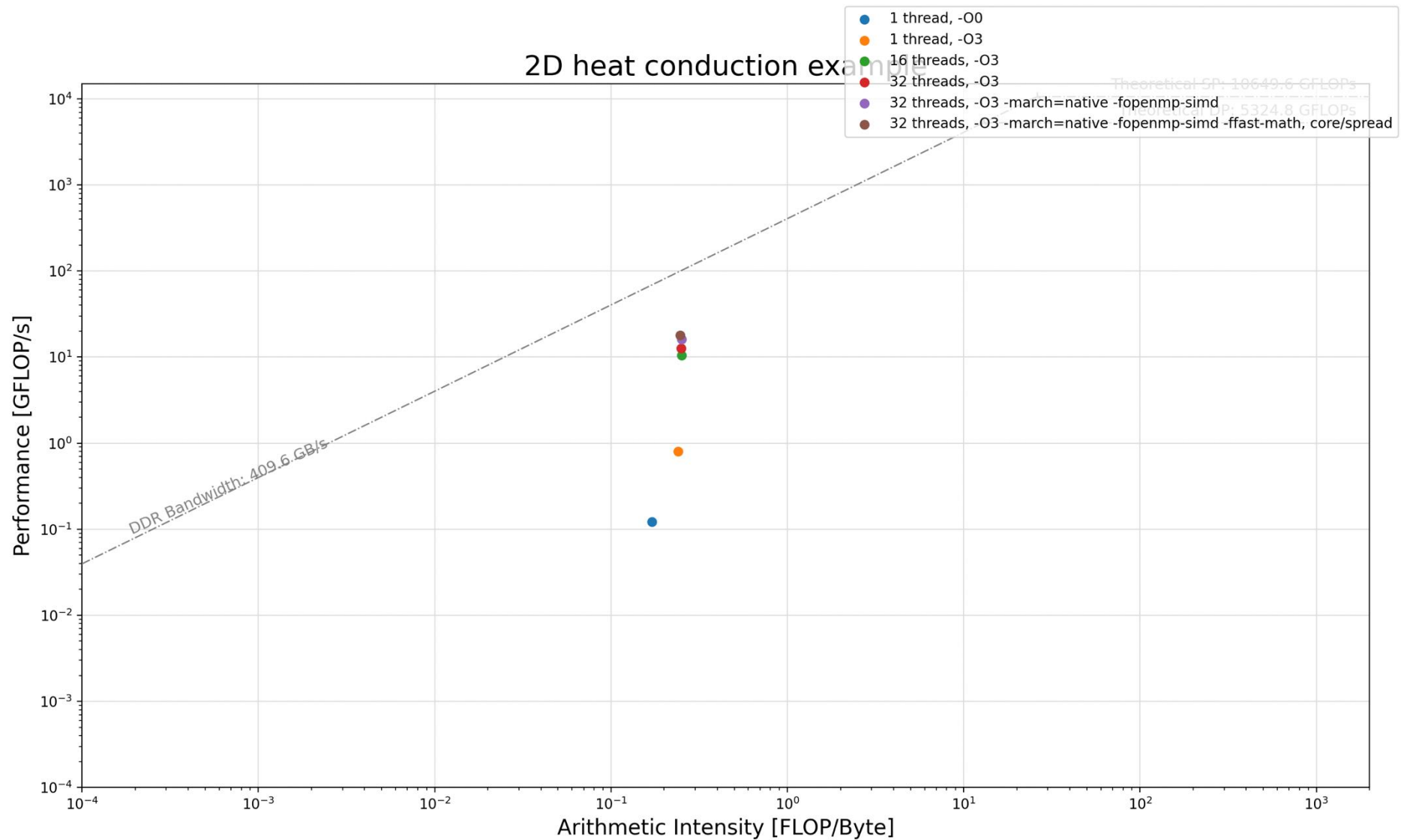
```
$ module purge  
$ module load 2024  
$ module load AMD-uProf/5.0.1479  
$ AMDuProfPcm roofline -X -o res.csv -- <prog> <args>
```

```
[Profile Data]  
Application,./a.out  
Type,DRAM  
Name,./a.out  
Throughput,6.7257  
Bandwidth,2.6767
```

Output example

$$AI = \frac{\text{Throughput}}{\text{Bandwidth}}$$


Roofline model



| How to estimate energy footprint?

- Add the following options to the job script and submit it:

```
#SBATCH --ear=on  
#SBATCH --ear-policy=monitoring  
  
export EARL_REPORT_LOOPS=1
```

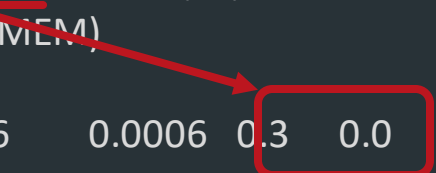
- Read more:
<https://servicedesk.surf.nl/wiki/spaces/WIKI/pages/62226671/Energy+Aware+Runtime+EAR>

| How to estimate energy footprint?

- EAR:

```
$ eacct -j <job-id>
```

JOB-STEP	AID	USER	APPLICATION	POLICY	NODES	AVG/DEF/IMC(GHz)	TIME(s)	POWER(W)					
GBS	CPI	ENERGY(J)	GFLOPS/W	IO(MBs)	MPI%	G-POW (T/U)	G-FREQ	G-UTIL(G/MEM)					
11041163-0	2747012												
maximm	job.sh		MO	1	2.52/2.60/1.47	56.64	253.62	1.83	0.44	14366	0.0006	0.3	0.0
0.00/---	---	---											



| SLURM



| SLURM

- **Nodes:**
 - **-N, --nodes** – request a number of nodes for a job
- **Tasks:**
 - **-n, --ntasks** – sbatch does not launch tasks, it requests an allocation of resources and submits a batch script. This option advises the Slurm controller that job steps run within the allocation will launch a maximum of *number* tasks and to provide for sufficient resources.
 - **--ntasks-per-node** - request that *ntasks* be invoked on each node.
- **CPUs**
 - **-c, --cpus-per-task** - Advise the Slurm controller that ensuing job steps will require *ncpus* number of processors per task. Without this option, the controller will just try to allocate one processor per task.

SLURM

- **Sockets**

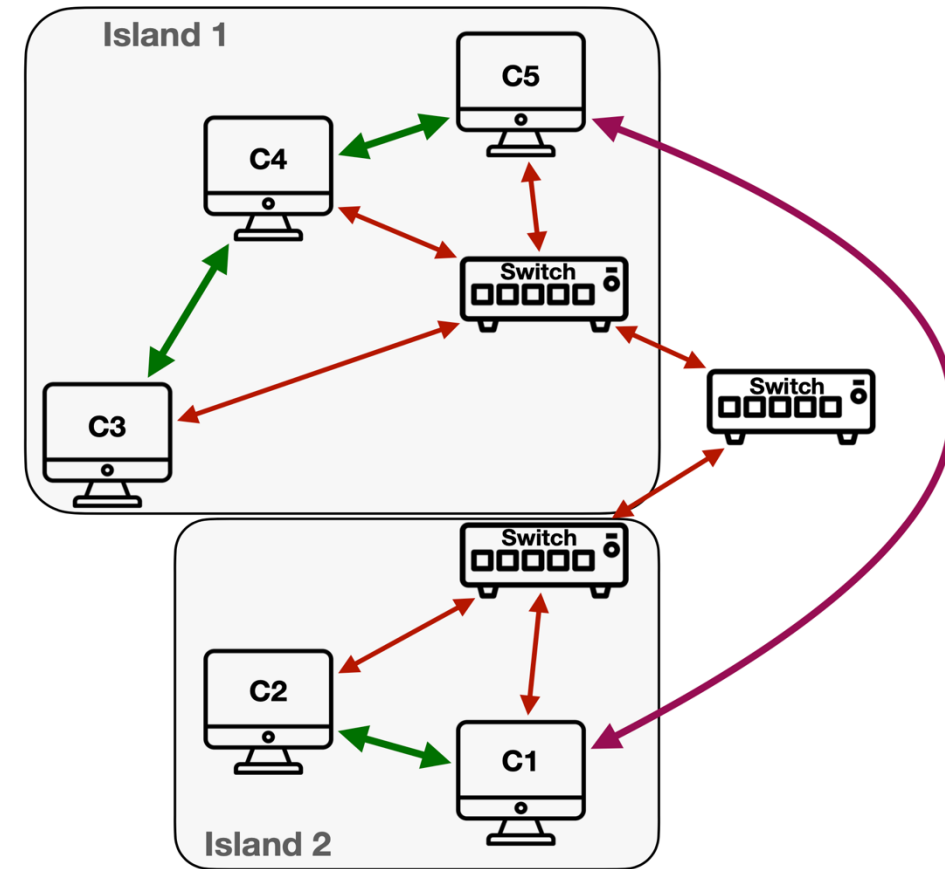
- **--sockets-per-node** - Restrict node selection to nodes with at least the specified number of sockets

- **GPUs**

- **--gpus-per-node** - Specify the number of GPUs required for the job on each node included in the job's resource allocation.

- **Allocations**

- **--contiguous** - If set, then the allocated nodes must form a contiguous set
- **--exclusive** - The job allocation can not share nodes with other running jobs



| SLURM

- **SLURM_JOB_NODELIST** - list of nodes allocated to the job.
- **SLURM_NTASKS** - set to value of the **--ntasks** option, if specified. Or, if either of the **--ntasks-per-node** or **--ntasks-per-gpu** options are specified, set to the number of tasks in the job.
- **SLURM_CPUS_PER_TASK** - number of cpus requested per task. Set if the **--cpus-per-task** option is specified.
- **SLURM_GPUS** - number of GPUs requested. Only set if the **-G, --gpus** option is specified.